A movable source-width certificate for the 0.874836 boundary

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Closed all-Dirichlet endpoint passed local Lean checking with only the three standard axioms. Clean packaged replay, canonical comparison and independent-kernel checks pending. No tracker admission is claimed.

Abstract

The locally checked Lean construction proves the all-moduli Dirichlet boundary $218709/250000 = 0.874836$. A new actual affine source line keeps the moving marked HIGH width, uses the established incumbent for LOW, and preserves a separate ERROR reserve. Three detector regimes and exact continuous row certificates carry this line through the weighted numerator and continuation. We distinguish the closed local theorem from the pending independent replay.

1 Statement and fixed geometry

The locally checked theorem is: for every positive modulus n, every Dirichlet character χ modulo n, and every $s \in \mathbb{C}$,

$$\Re s > \vartheta, \qquad \neg(\chi = 1 \text{ and } s = 1) \quad \Longrightarrow \quad L(\chi, s) \neq 0. \tag{1}$$

The OpenAI–Collet framework reduces (1) to improving the Eisenstein Hecke-family global zero supremum β [1, 3]. We use the accepted $\beta \le 21871/25000$ as an incumbent input, never the new conclusion $\beta \le \vartheta$.

The new rational geometry is

$$\begin{aligned}
h &= \frac{3181}{4000}, & d_- &= \frac{15893}{20000}, & d_+ &= h + 10^{-10},\\
\ell &= \frac{11}{3} - 4\vartheta = \frac{31373}{187500}, & x &= 1 + \ell - h = \frac{558109}{1500000}, & y &= h - 2\ell = \frac{690907}{1500000},\\
\kappa_* &= 2\vartheta - 1 = \frac{93709}{125000}, & \lambda &= \frac{19793}{5000}, & B &= \frac{987047}{1000000}.
\end{aligned}$$

The actual initial affine moment is proved on

$$\frac{41}{100} \le m_1, m_2 \le \frac{11}{25}, \qquad \frac{17}{500} \le b \le \frac{1}{25}, \qquad m_1 + m_2 + \lambda b \le B. \tag{2}$$

For equal lengths this is $2m + \lambda b \le B$. The literal energy, masks, selected primes, seminorms and height losses are preserved; the constants precede the moving row and original slot subset.

2 LOW: use the proved incumbent without circular feedback

Put $q = 839/1250$, $\kappa_0 = 9371/12500$, and $r_L = 1/4000000$. Here $\kappa_0 = 2(0.874840) - 1$ is fixed by the predecessor. For a cell upper endpoint b, set

$$\begin{aligned}
D(b) &= B - 1 + (6\kappa_0 - \lambda)b,\\
a(b) &= \frac{q}{6(\kappa_0 - q)} D(b) + \frac{r_L}{1 - q/\kappa_0},\\
A(b) &= B - (\lambda - 1)(b - 10^{-5}).
\end{aligned} \tag{3}$$

The exact identity

$$a(b) - \frac{q}{6\kappa_0}\big(D(b) + 6a(b)\big) = r_L \tag{4}$$

pays the complete LOW removal cost: both the capacity and its removal price change with the incumbent. The accepted positive/zero energy pair supplies all heights and masks. The stronger target is weakened to Collet's generic LOW fee ledger for the outer contraction.

3 HIGH: retain actual width and the common-allocation geometry

The marked HIGH source reuses timbalud's asymmetric inverse/plain counting and common-radical architecture [4]. The actual first numerator also satisfies a valuation bound whose logarithmic form is $4L_c \le 5(c+d)$. Together with the original radical and cancellation gates, it gives a common-width rate

$$\rho(\delta) = 1 - \frac{10\delta}{3(5-2\delta)}, \qquad \gamma(\delta) = \delta - \max\left(4\rho(\delta) - 1, \frac{2 + 22\rho(\delta)}{21}\right). \tag{5}$$

The second allocation pays rate δ. Neither reflection orientation changes the first-numerator identity or substitutes a synthetic source norm.

Write $u, f, v \ge 0$ for the removed common, frozen and surviving widths. The two actual allocation stages and the tiny-slot discard prove

$$u + f + v \ge S(b) := b - 10^{-5} - 10^{-3}, \qquad Q \ge F(b) - 2u, \quad F(b) = \frac{122}{125}\big(1 - 6a(b) - 10^{-3}\big). \tag{6}$$

The crucial change is the moving quantity $S(b)$: the original selected width is retained through both allocations, before choosing the cell. Replacing it everywhere by the smallest permitted width loses useful payment at larger cells, where $a(b)$ and $A(b)$ also move.

Set

$$r(\delta) = \frac{32}{39}\frac{\delta(5-6\delta)}{5-2\delta}, \qquad t(\delta) = \frac{6\rho(\delta) - 1}{250}, \qquad \tau_s = \frac{13}{400}.$$

The marked payment is $\gamma u + \delta f + r\min(v, \tau_s Q)$; its large sixth-power tail uses $\gamma u + \delta f + tQ$. If $v \le \tau_s Q$, compare the three width rates and use the first inequality in (6). If $v > \tau_s Q$, insert the second inequality in (6) and use $2r\tau_s \le \gamma$. The analogous tail step uses $2t \le \gamma$. Consequently it suffices to check four scalar channels. With

$$H(b, \delta) = 1 - \rho(\delta) - \delta A(b) - a(b)(5 - 6\rho(\delta)),$$

we display a conservative common ledger charging $1/20000$ and a separate $1/200000$ raw-root allowance:

$$H + \gamma S - \tfrac{1}{20000} - \tfrac{1}{200000} \ge 10^{-6}, \qquad H + rS - \tfrac{1}{20000} - \tfrac{1}{200000} \ge 10^{-6},$$
$$H + r\tau_s F - \tfrac{1}{20000} - \tfrac{1}{200000} \ge 10^{-6}, \qquad H + tF - \tfrac{1}{20000} - \tfrac{1}{200000} \ge 10^{-6}.$$

The marked channels reserve the latter charge before the raw-root estimate; the unmarked common-width channel has the stronger bound without that charge. These hold on the full moving cell range $3401/100000 \le b \le 1/25$ and $q \le \delta \le 7/10$. Monotonicity pays $7/10 \le \delta \le 3/4$ directly. The actual selected-family, physical-fiber and all-height suppliers preserve their original coefficients, frequency collisions and finite dictionaries.

4 ERROR: preserve the separate deletion capacity

The improved LOW incumbent does not change the original error-deletion capacity. Its defect is

$$D_E(b) = B - 1 + (9/2 - \lambda)b, \tag{7}$$

with $9/2 = 6(3/4)$. In particular, the first error branch has the independent uniform reserve

$$\frac{a(b)}{4} - \frac{D_E(b)}{6} \ge \frac{1}{1000}. \tag{8}$$

The sixth and seventh branches have separately proved positive reserves; all three exceed the pre-loss fee-ledger reserve $7/30000000$. Absorbing branch losses halves this to $7/60000000$ in the complete normalized physical sum: its saving term is $-M(7/60000000)$, with M the source scale and allocation/shell losses still displayed separately. The source proof includes the full retained radical witness and the moving-prime average exactly once. The normalized ERROR family is converted to physical paid terms before it joins the actual MAIN family. Zero and tail terms use the new $a(b), A(b)$ as well: for example, $1 - A(b) - 5a(b) \ge 227876729/3446640000$ and the full amplifier coefficient is at most 2048. Thus a main-only scalar improvement is not being passed off as a complete source improvement.

5 Use the new moment inside the actual detector count

DaysSky's construction applies an initial affine moment separately to each actual detector dictionary character [5]. We keep this order. For inverse length r, plain length m, cutoff t with $r + m = t$, zero density δ and normalized amplitude p, the three count inputs are

$$\begin{aligned} I &= 1 - \delta\big(p + (1-p)r\big), \\ P &= 1 - 2\delta m - 2\delta p\frac{B - 2m}{\lambda}, \\ L &= 1 - \delta + (5/6 - \delta)(t-1). \end{aligned}$$

Balancing these gives, with positive denominators,

$$\begin{aligned} D(p) &= 3 - p - 4p/\lambda, & a_p &= 2 - 4p/\lambda, & c_p &= 1 - 2pB/\lambda, \\ J(\delta,p) &= (5/6-\delta)D(p) + \delta(1-p)a_p, \\ t_N &= 1 + \frac{\delta(1-p)c_p}{J(\delta,p)}, & R_{\mathrm{aff}} &= 1 - \delta + \frac{(5/6-\delta)\delta(1-p)c_p}{J(\delta,p)}. \end{aligned} \tag{9}$$

The selected width must stay in the proved window of (2), including the rounding subtraction. The detector checks this with $m \le 847/2000$ and $1 \le t_N \le 3/2$. Smaller widths or inadmissible plain lengths retain the original inverse/plain estimates. The global count uses three regimes:

Regime	Density interval	Amplitude interval
New affine line	$[451/1000, 49/100]$	$[49/100, 1/2]$
Accepted affine fallback	$[89/200, 451/1000]$	$[199/400, 1/2]$
Original count	complementary region	$[0, 1/2]$

The shared inverse/prime moment constants and balanced count parameters are fixed once. The new line also supplies the accepted fallback window, so there is no independent unproved analytic assumption on that rectangle. The amplitude-class theorem consumes this same actual count.

The finite detector dictionary is formed after the final target character is fixed. Its moment degrees are maximized and its finite seminorm sets united before selecting a common small height exponent for its labels. This is uniformity over a finite dictionary; it does not assert one moment degree for infinitely many target characters.

6 Weighted numerator and whole-domain certificates

Akash Levy's weighted-numerator architecture and Collet's full continuation transport are retained [2, 3]. The source line changes the available numerator capacity on the upper height interval to

$$C_N(d) = \frac{2y - (2-B)d}{\lambda}.$$

On the lower interval it remains $C_O(d) = \min(d, 2y-d)/(6\kappa_*)$. The physical shifted capacity is used with switch $t_0 = 103/100000$; this is distinct from the source frequency allowance 10^{-3}. The choice pays the literal large-shift gate and preserves the unclipped physical slot cap $37/1000$.

For either capacity C the normalized critical row is

$$E(d, C, \delta, p) = \frac{1+\delta}{2} + \frac{13h}{75} - \frac{y}{2} - \frac{\ell}{2} + \ell\delta p + d\left(\frac{1}{4} - \frac{17}{50}\right) - \frac{\delta p C}{2} + \frac{3d}{4}R(\kappa_*, \delta, p). \tag{10}$$

Exact rational Bernstein certificates prove $E \le \vartheta - 10^{-7}$ over all $0 \le \delta \le 3/4$, $0 \le p \le 1/2$. They cover the three count regimes, not just a sampled critical point. Affinity in d reduces the two full height intervals to five ends: $d_0 = 1/200$, y, d_- with C_O, and d_-, d_+ with C_N. Positivity of the count denominators, source geometry, selected-width and tail gates, amplitude monotonicity and the varying $\kappa = 2\beta - 1$ feedback are proved separately and consumed by the row provider.

7 Actual assembly, verification scope and attribution

The source, detector and numerator components are joined through the following literal contracts:

$$\begin{aligned} &\text{actual LOW/HIGH/ERROR} \longrightarrow \text{full geometry budget} \longrightarrow \text{initial affine moment} \\ &\longrightarrow \text{dictionary moment/count} \longrightarrow \text{finite weighted numerator} \longrightarrow \text{common-probe continuation.} \end{aligned} \tag{11}$$

The full geometry budget includes both reflections, shifted Mellin heights, zero/tail terms and the original retained ERROR witness. The initial moment has the quantifiers of the original source theorem, with the new numerical line. The finite numerator uses the new physical lengths and the actual source and count receipts. Both lower and upper estimates are for the same normalized probe; the inherited correction factor is analytic and nonzero on the required new half-plane.

The continuation argument keeps β as the global supremum. Under a contradiction assumption $\beta > \vartheta$, the common positive continuation reserve is chosen before the primitive target characters. Target-dependent heights may then be chosen, including the finite dictionary maximum above. The resulting common gap improves the global supremum. This quantifier order is essential: a target-by-target gap without a common positive reserve would not establish the supremum contradiction.

Verification status. Closed all-Dirichlet endpoint passed local Lean checking with only the three standard axioms. Clean packaged replay, canonical comparison and independent-kernel checks pending. No tracker admission is claimed. The changed private Lean modules have per-module compiler receipts and reports restricted to `propext`, `Classical.choice` and `Quot.sound`. The continuous scalar certificate additionally has an independent exact rational verifier. These checks are distinct from a clean replay of the complete packaged dependency graph, canonical-target comparison, independent kernel acceptance and tracker admission. The endpoint and replay status is recorded separately; this manuscript does not turn component receipts into an unperformed verification claim. Accepted predecessor sources remain unchanged.

What changes, and what is inherited. The new design keeps the moving HIGH supply, the literal first-numerator norm facet and the accepted LOW incumbent together when choosing the affine line. It carries separate ERROR reserves and all three detector regimes through the existing source and weighted-numerator consumers. It changes the height geometry and regenerates the continuous endpoint certificates. The analytic continuation architecture, finite-source decomposition, moment estimates and large-scale counting mechanisms are inherited and explicitly reused.

Attribution. OpenAI supplies the protected 7/8 framework and core library. Hailey Collet supplies Cycle25/Cycle61, the initial affine moment, physical source decomposition and continuation architecture. Akash Levy supplies the weighted-numerator contribution and its inherited credits. Timbalud supplies the asymmetric marked HIGH counting and common-radical source mechanism. DaysSky supplies the actual affine detector moment and finite-dictionary height order. The accepted 0.874840 integration is the fixed incumbent input. Related count-side affine arithmetic by liqsweep/blueberryvertigo is acknowledged, without using its proposed endpoint as a proved input. Zhongpai Gao, with AI assistance, supplies this source-line design, the new numerical geometry, exact certificates and the private fresh source/count/numerator integration. Existing source notices and predecessor licenses are retained; this manuscript is released under Apache-2.0 subject to those notices.

References

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